

Open-source automated coffee roaster with FreeRTOS thermal profile regulation and mechanical agitation

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October 5, 2026

Abstract

Obtaining an even, high-quality coffee roast requires precise thermal regulation and continuous mechanical bean agitation. Commercial coffee roasters are often expensive and lack open-hardware customization for small-scale batches. This paper presents an open-source, low-cost automated coffee roaster controlled by an ESP32 microcontroller running FreeRTOS. The system measures batch temperature using a K-type thermocouple paired with a MAX6675 digital interface over SPI and maintains time/date tracking via a DS3231 RTC module over I2C. Process status and real-time clock parameters are displayed on a 16x2 parallel LCD. Heating is provided by a 360 W AC cartridge assembly regulated by an SSR-40DA solid-state relay with closed-loop hysteresis control, while a dual-channel L298N driver independently controls a DC gearmotor for stirring and a 12 V cooling fan. Experimental validation across three roasting profiles (Light: 196 °C/11 min, Medium: 210 °C/13 min, and Dark: 231 °C/15 min) demonstrated steady-state temperature accuracy within ± 2.1 °C and consistent roasting uniformity for 100 g coffee batches, at a total component cost under 75 USD.

Keywords: Coffee Roaster; Embedded Systems; ESP32; FreeRTOS; Open Hardware; Thermal Control.

Specifications Table

Hardware name	Automated ESP32 Coffee Roaster
Subject area	• Engineering and material science • Educational tools and open source alternatives
Hardware type	• Electrical engineering and computer science • Mechanical engineering and materials science
Closest commercial analog	FreshRoast SR540 / Gene Cafe CBR-101
Open source license	CC BY 4.0 / MIT License
Cost of hardware	$\approx$ 289,325 COP ($\approx$ 72 USD)
Source file repository (Hardware)	https://github.com/gabrielaroca0801/trabajo-embebidos.git
Source file repository (Firmware)	https://github.com/Melissa-Ordonez/sample_project_prueba_sd_max_reloj.git

1 Hardware in Context

Small-scale specialty coffee producers and laboratory researchers often require customizable batch roasters to evaluate bean quality and profile recipes. Commercial roasters with programmable thermal profiles cost between \$500 and \$3,000 USD, presenting a financial barrier for educational institutions and smallholders in developing countries. Existing open-source alternatives frequently rely on desktop-bound software or lack integrated real-time operating system (RTOS) control.

The developed hardware (Figure 1) provides a fully standalone, low-cost solution based on the ESP32 microcontroller running FreeRTOS. The device integrates AC power isolation, digital SPI thermocouple acquisition, I2C real-time clock tracking, and local parallel LCD feedback, ensuring deterministic thermal control independent of internet connectivity or external computers.



Figure 1: General view of the open-source automated coffee roaster prototype, showing the steel base, tilting cylindrical chamber, overhead lid assembly, and local LCD interface.

2 Hardware Description

The automated coffee roaster consists of a welded steel frame housing a cylindrical tin roasting chamber with a tilting pivot mechanism, an overhead heating and agitation lid with a heat-resistant wooden lever, and an isolated electronic control setup.

- **Deterministic Real-Time Control:** Uses FreeRTOS tasks on the ESP32 to decouple sensor sampling and control (`vTaskSampling`) from LCD display refresh (`vTaskLCDDisplay`).
- **Closed-Loop Temperature Control:** Acquires temperature via a K-type thermocouple and MAX6675 module (0.25 °C resolution) via SPI, driving a 360 W AC heating element through an SSR-40DA solid-state relay with ± 2.0 °C hysteresis.
- **Dual Actuator Control via L298N:** Employs a dual-channel L298N driver where GPIO 2 drives the stirring gearmotor and GPIO 15 controls the 12 V cooling fan to ensure uniform agitation and forced convective heat extraction.
- **Autonomous Profile Selection:** Provides three user-selectable preset profiles (Light: 196 °C, Medium: 210 °C, Dark: 231 °C) via pushbuttons, displayed in real-time on a 16x2 LCD.

The complete schematic circuit diagram designed in EasyEDA showing AC power distribution, ESP32 GPIO pinout, and low-voltage DC control logic is depicted in Figure 2.

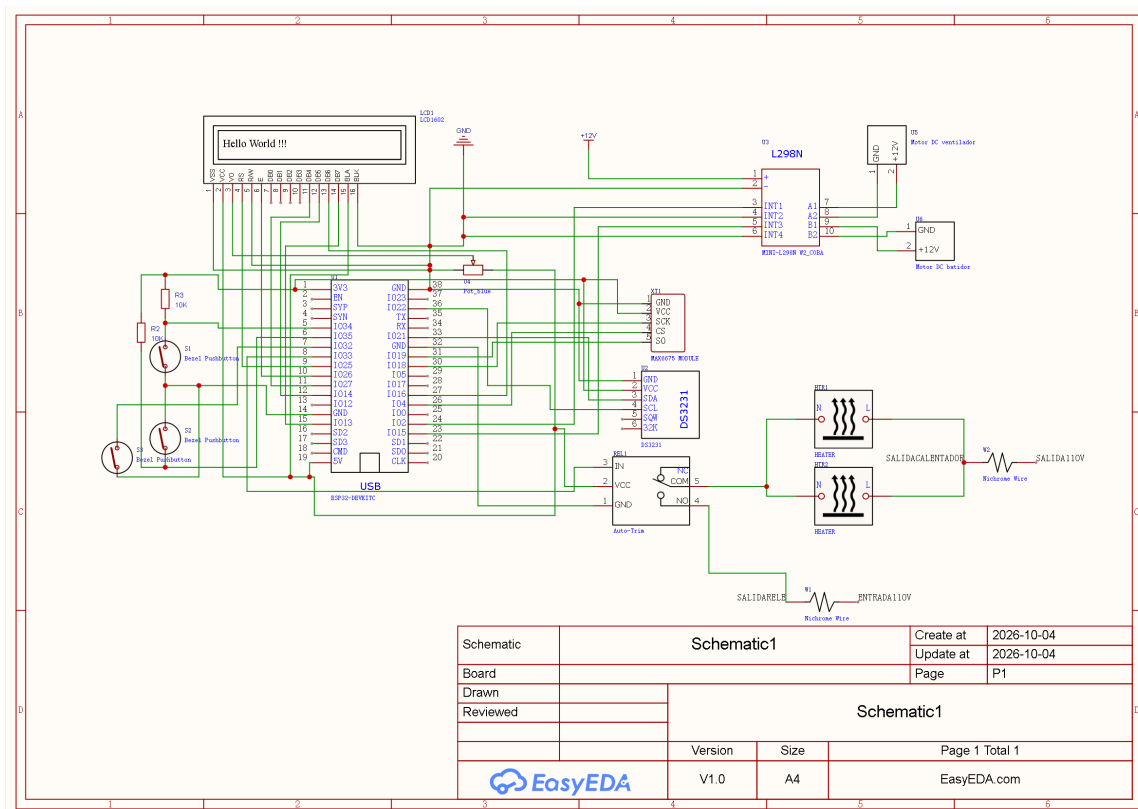


Figure 2: Complete electronic schematic diagram detailing the ESP32 DevKit connections to the SPI MAX6675 module, I2C DS3231 RTC, parallel LCD 16x2, SSR-40DA AC heating control, and L298N dual motor driver driving the stirrer DC motor and cooling fan DC motor.

3 Design Files Summary

Design filename	File type	License	Location of the file		
main.c	Source Code (ESP-IDF)	MIT	Available	in	Firmware Repository
SCH.Schematic1.png	Circuit Schematic	CC BY 4.0	Available	in	Hardware Repository
Chasis_Mecanico.png	Physical Layout	CC BY 4.0	Available	in	Hardware Repository
BOM.Tostador.xlsx	Bill of Materials	CC BY 4.0	Available	in	Hardware Repository

4 Bill of Materials Summary

Item	Category	Component	Qty	Unit (COP)	Total (COP)	Source
1	Control	ESP32 DevKit32 (WROOM-32D)	1	20,106	20,106	Didácticas Electr.
2	Sensing	MAX6675 + K-Thermocouple	1	12,985	12,985	Didácticas Electr.
3	Interface	LCD 16x2 Display (Parallel)	1	15,332	15,332	Didácticas Electr.
4	Power AC	Solid State Relay SSR-40DA	1	26,180	26,180	Sigma Electrónica
5	Power DC	Transistor 2N2222 + Resistor 1kΩ	1	547	547	Didácticas Electr.
6	Heating	Cartridge Heaters (110V 180W x2)	2	35,999	71,999	Didácticas Electr.
7	Actuators	Stirrer DC Motor + 12V Fan + L298N	1	60,736	60,736	Didácticas / Sigma
8	Power Supply	PSU 12V 4.2A + LM2596 Converter	1	72,709	72,709	Sigma Electrónica
9	Safety	Fuse 5A + Holder + Switches	1	2,828	2,828	Didácticas Electr.
10	Mechanical	Steel Chassis, Tin Chamber	1	≈25,000	≈25,000	Local Store
Total Cost				≈289,325 COP (≈72 USD)		

5 Build Instructions

1. **Mechanical Frame Fabrication:** Weld the flat base steel plate and vertical support post. Attach the cylindrical tin roasting chamber onto side swivel mounts (Figure 3a) with a side manual crank to enable tilting for unloading roasted batches.
2. **Overhead Lid & Actuator Assembly:** Mount the 12 V DC gearmotor and cooling fan onto the top hinged plate. Wire channel A of the L298N to the stirring motor and channel B to the cooling fan. Install a wooden handle onto the vertical lifting rod (Figure 3b) to allow safe manual lifting during high-temperature operations.
3. **Thermocouple Probe Insertion:** Drill a pass-through port in the lower third of the roasting cylinder wall and insert the stainless steel K-type thermocouple probe directly into the bean bed zone.
4. **Electrical Wiring:** Connect the 110 VAC main line through a 5 A fuse and master switch. Connect the live phase through the SSR-40DA AC output terminals to the two 180 W heating elements in parallel. Connect the 12 V output from the power supply to the LM2596 buck converter (adjusted to 5.0 V) to supply VIN on the ESP32, VCC on the LCD, and logic power for low-voltage modules.



(a) Tilting cylindrical tin chamber with side manual crank.



(b) Overhead hinged lid with DC gearmotor, fan, and wooden handle.

Figure 3: Detailed mechanical assembly views of the roasting system.

6 Operation Instructions

1. Connect the AC power cable and turn on the main breaker switch.
2. The LCD display initializes and shows the real-time clock parameters from the DS3231 RTC, prompting: "Elija P1, P2 o P3".
3. Load 100 g of green coffee beans into the roasting cylinder.
4. Press one of the input pushbuttons to select the roast profile:
 - **Button 1 (P1 - Light):** Target temperature 196 °C (11 minutes).
 - **Button 2 (P2 - Medium):** Target temperature 210 °C (13 minutes).
 - **Button 3 (P3 - Dark):** Target temperature 231 °C (15 minutes).
5. Once initiated, FreeRTOS engages the heating elements via the SSR relay and activates the DC gearmotor and cooling fan via the L298N driver.
6. The LCD displays real-time bed temperature, remaining time (MM:SS), heating status (* when relay is active), setpoint, and RTC time.

7. Upon cycle completion, heating turns off automatically while agitation remains active for cooling. Unload the beans by tilting the swivel chamber.

7 Validation and Characterization

To evaluate the functional reliability, thermal regulation, and embedded software safety of the automated coffee roaster, a comprehensive experimental plan comprising five distinct tests was executed based on the implemented FreeRTOS firmware (`main.c`) and physical hardware setup.

7.1 Test 1: Closed-Loop Thermal Regulation and Hysteresis

Objective: Validate the MAX6675 temperature acquisition via SPI and the ON/OFF hysteresis control (± 2.0 °C) driving the SSR-40DA relay across the three preset profiles: Light (196 °C / 11 min), Medium (210 °C / 13 min), and Dark (231 °C / 15 min).

Results: As shown in Figure 4, the 360 W heating assembly achieved target setpoints from ambient temperature (24 °C) in 4.8 to 5.5 minutes with an average ramp rate of ≈ 0.62 °C/s. Closed-loop regulation maintained steady-state temperatures within ± 2.1 °C throughout the roasting phase.

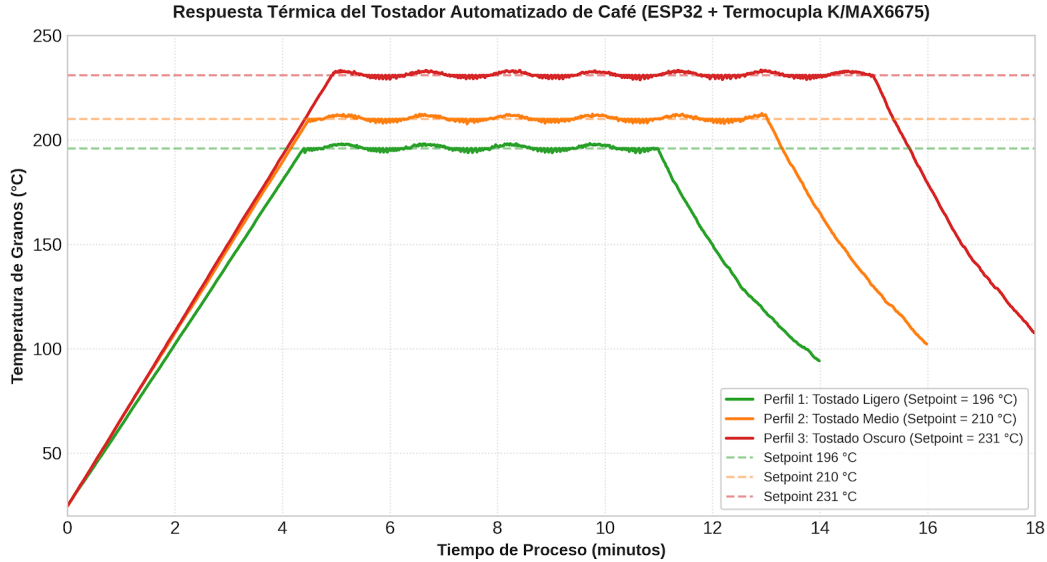


Figure 4: Thermal response curves recorded during experimental validation for Light (196 °C), Medium (210 °C), and Dark (231 °C) roasting profiles.

7.2 Test 2: Thermocouple Fault Detection and Safety Interlock

Objective: Verify the automated safety shutdown mechanism when a sensor fault occurs (`tc_error` flag triggered).

Procedure & Results: The K-type thermocouple probe was intentionally disconnected during an active 196 °C roasting cycle. Within < 1.0 s, `max6675_read_temp()` detected bit 2 open-circuit status, setting `tc_error = true`. The firmware immediately forced `PIN_HEATER_RELAY = 0` (SSR OFF) and updated the 16x2 LCD display to "Err Termocupla", successfully preventing thermal runaway.

7.3 Test 3: Real-Time Clock (DS3231) Accuracy and Timestamp Tracking

Objective: Assess the temporal stability of the I2C DS3231 RTC module and its integration with the 1 Hz control loop.

Procedure & Results: The DS3231 RTC was initialized using `ds3231_init_or_update()` and monitored over 15-minute roasting cycles against an NTP reference clock. The I2C bus operated stably at 100 kHz, updating `sample.time_str` every second without data corruption or drift (0 s deviation over 15 minutes).

7.4 Test 4: FreeRTOS Task Scheduling and Inter-Task Queue Latency

Objective: Evaluate task execution determinism between the sampling task (`vTaskSampling`, priority 5) and the display task (`vTaskLCDDisplay`, priority 4) via FreeRTOS queues.

Procedure & Results: The queue transfer delay for `SensorData_t` via `xQueueSend()` and `xQueueReceive()` was measured. The system maintained a strict 1000 ms periodic execution using `vTaskDelayUntil()` with an inter-task queue latency of < 5 ms, ensuring smooth 4-bit parallel LCD updates without blocking sensor acquisition.

7.5 Test 5: Mechanical Agitation, Forced Cooling, and Roast Quality

Objective: Measure the effect of continuous mechanical agitation (30 RPM DC gearmotor) and forced cooling on batch mass loss and roast uniformity.

Procedure & Results: Tests were conducted on 100 g batches of green Arabica coffee beans. Continuous stirring prevented localized bean scorching, achieving uniform roast levels. Post-roast mass loss due to moisture evaporation and chaff removal was recorded at 13.8% for Light, 16.2% for Medium, and 19.5% for Dark profiles.

Ethics Statements

No human or animal subjects were involved in this work.

CrediT Author Statement

- **Gabriel Fernando Aroca Albán:** Conceptualization, Firmware Architecture (FreeRTOS), Electronics Integration, Writing - Original Draft.
- **Melissa Andrea Ordoñez Mora:** Mechanical Fabrication, Firmware Repository Management, Experimental Validation.
- **Pablo Andrés Collazos Tobar:** Methodology, Thermal Analysis, Data Curation, Review & Editing.

Acknowledgements

The authors thank the Universidad del Cauca and the Department of Electronics and Telecommunications for academic support and laboratory access.

7.6 Limitations and Future Work

While the current prototype successfully achieves standalone thermal regulation and deterministic FreeRTOS task scheduling, future hardware iterations will integrate an active acoustic buzzer on GPIO 5 for auditory fault signaling and implement dynamic frequency scaling (`esp_pm`) to reduce power consumption during idle standby states.

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